Lab 1 Link-layer

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3.2 Writing Task 1 (WT1)

1. There are 827 frames in the filtered results out of a total of 1729 frames.
2. The destination address is `ff:ff:ff:ff:ff:ff` which is special as it is the Ethernet broadcast MAC address. This sends the frame to all hosts on the same local Ethernet broadcast domain.
3. The 71st byte of this frame is:
   `0x15`

Checkpoints

Checkpoint 1

```
lab-netstack-premium/lab1 on  lab1 [✘!?⇡] took 2s
❯ sudo ./test_device en0
Detected interfaces:
- ap1
- en0
- awdl0
- utun0
- llw0
- utun1
- utun2
- utun3
- utun4
- utun5
- utun1024
- lo0
- anpi0
- anpi3
- anpi1
- en4
- en5
- en6
- en1
- en2
- en3
- bridge0
- gif0
- stf0
- nan0
Added en0 as device ID 0.
findDevice and duplicate add both returned 0.
```

The program used `libpcap` to find and list all host interfaces (via `pcap_findalldevs`) then added `en0` as a device assigning it ID 0. Calling `findDevice` returned the device ID (0 in this case). Calling `addDevice` on `en0` also returned 0, for the device ID instead of 1. This means `addDevice` is an idempotent operation.

Checkpoint 2

In Terminal 1:

```
lab-netstack-premium/lab1 on  lab1 [✘!?⇡]
❯ make test_packetio
cc  -pthread -o test_packetio tests/test_packetio.o src/device.o src/packetio.o -lpcap

lab-netstack-premium/lab1 on  lab1 [✘!?⇡]
❯ sudo ./test_packetio en0
Sent a test frame on en0 with EtherType 0x88B5.
Captured the test frame on device ID 0.

lab-netstack-premium/lab1 on  lab1 [✘!?⇡]
❯
```

In Terminal 2:

```
lab-netstack-premium on  lab1 [✘+?⇡]
❯ sudo tcpdump -i en0 -e -XX ether proto 0x88b5
Password:
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on en0, link-type EN10MB (Ethernet), snapshot length 524288 bytes
16:17:09.699289 96:eb:48:95:4b:7d (oui Unknown) > Broadcast, ethertype Unknown (0x88b5), length 31
:
        0x0000:  ffff ffff ffff 96eb 4895 4b7d 88b5 7061  ........H.K}..pa
        0x0010:  636b 6574 696f 2050 5432 2074 6573 74    cketio.PT2.test

```

The program used `pcap_sendpacket()` to inject an Ethernet II frame with payload `packetio PT2 test` and EtherType `0x88B5`; its header is shown above. A `pcap_loop` capture thread received the frame and forwarded it to the registered callback (in this case called `handleIncomingFrame`).

Tcpdump confirms that the injected frame contains: (see typescript file for more)

```
Destination MAC: ff:ff:ff:ff:ff:ff
Source MAC: 96:eb:48:95:4b:7d
EtherType: 0x88B5
Payload: packetio PT2 test
```

Code Arrangement

PT1

- `device.h`
- `device.c`

PT2

- `packetio.h`
- `packetio.c`

Tests

- `test_device.c`
- `test_packetio.c`

Other

- `Makefile`